

SANDEEP NAUTIYAL

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EDUCATION

Dronacharya College of Engineering Gurugram, India
B.Tech, Computer Science — Cyber Security & IoT (incl. Blockchain) **CGPA: 7.7 / 10** 2024 – 2028 (Expected)
VIDYA School Gurugram, India
Class XII, Science Stream | **74%** 2023

TECHNICAL SKILLS

Languages: JavaScript, Python, C++, Java, C, SQL
Web & Backend: React.js, Node.js, Express.js, MongoDB (MERN Stack), Socket.io, REST APIs, TailwindCSS, HTML5/CSS3
Cloud & DevOps: AWS (EC2, S3, VPC, SNS, Route 53, EBS), GitHub Actions, CI/CD Pipelines, Linux Administration, Vercel, Railway, Bash Scripting
Security & Systems: AES-256/GCM Encryption, PBKDF2 Key Derivation, Zero-Knowledge Architecture, JWT, Rate Limiting, VPC Security Groups, TCP/UDP Networking, Cryptographic Protocol Design
Tools: Git, GitHub, VS Code, Object-Oriented Programming, Data Structures & Algorithms

PROJECTS

AbyssChat | *React, Vite, Node.js, Socket.io, AES-GCM, GitHub Actions, Vercel, Railway* 

- Architected a zero-knowledge, real-time chat application with ephemeral chatrooms ensuring no message retention or user tracking, eliminating server-side data exposure by design.
- Engineered end-to-end encryption using AES-GCM (256-bit) with 100,000 PBKDF2 iterations for key derivation, meeting FinTech-grade cryptographic standards comparable to production messaging systems.
- Built a full CI/CD pipeline via GitHub Actions automating deployments to Vercel (frontend) and Railway (backend); integrated rate limiting to neutralise DDoS and brute-force attack vectors.

SayCaster | *Python, PyGame, Custom Rendering Engine*

- Engineered a pseudo-3D raycasting engine entirely from scratch in Python, rendering real-time environments from 2D map arrays with zero external game engine dependencies.
- Implemented fisheye correction, angle normalisation, and collision detection — demonstrating low-level systems thinking and strong algorithmic problem-solving.

AWS Cloud-Hosted Gaming Infrastructure | *AWS (EC2, VPC, EBS), Linux (Ubuntu), Bash*

- Designed and deployed a persistent multiplayer server on AWS EC2, optimising instance selection and EBS storage for cost-performance efficiency, achieving **99.9% uptime** under concurrent load.
- Configured VPC security groups with fine-grained TCP/UDP ingress and egress rules for low-latency access; automated infrastructure tuning via Bash scripts on Linux CLI.

CERTIFICATIONS

• **OCI Certified Foundations Associate** — Oracle
• **Java** — HackerRank
• **Frontend Developer (React)** — HackerRank
• **Python** — HackerRank